

CHEMISTRY: CHAPTER 11

ENVIRONMENTAL CHEMISTRY: AIR | QUICK PREP

Ye chapter abhi class mein nahin parhaya gaya. Paper 24 ke 35 marks isi chapter se hain, is liye paper se pehle ye sheet dhyan se parho. Baqi 15 marks Chapter 10 aur 8 ki revision hain.

1. Hawa kis cheez se bani hai

Saaf, khushk hawa mein taqreeban 78% **nitrogen** aur 21% **oxygen** hai. Baqi thori si **noble gases** aur **carbon dioxide**.

Zameen ke gird gases aur paani ke bukharaat (water vapour) ke parde ko **atmosphere** kehte hain.

Pollutant: "Anything that is in the air, water or soil which has a harmful effect on some part of the environment."

Yani jo cheez hawa, paani ya mitti mein ho kar mahaul ko nuqsan de.

2. Chhay bade air pollutants (Fig. 11.2)

Pollutant	Kahan se aata hai	Nuqsan
SO ₂ (sulphur dioxide)	Fossil fuels mein sulphur hota hai; power plants, gaariyan, factories	Asthma, bronchitis barhata hai; acid rain aur haze
NO, NO ₂ (nitrogen oxides, NO _x)	Gaariyon ke engines, power plants; bijli (lightning)	Acid rain, photochemical smog, sar dard, saans ki bimari
CO ₂ (carbon dioxide)	Fossil fuels jalana, jangal kaatna (deforestation)	Global warming aur climate change
CO (carbon monoxide)	Incomplete combustion , jab oxygen kam ho	Zehreeli gas (neechay dekho)
CH ₄ (methane)	Rice fields , wetlands, landfills; gaaye, bakri, bhair ka hazma (belching)	Global warming
Ground-level ozone (O ₃)	NO _x + VOCs, dhoop mein	Saans ki bimari, asthma; fasal aur jangal ko nuqsan

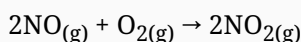
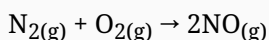
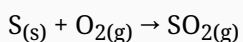
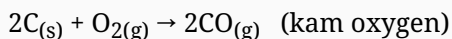
Particulate matter: hawa mein bohat barik zarraat (soot, dhuan, dust). Ye **phephron (lungs) mein gehrai tak** chale jate hain: saans ki bimari aur **lung cancer**.

Carbon monoxide khoon ke **haemoglobin** se jur kar **carboxyhaemoglobin** banati hai, jo oxygen nahin utha sakta. Is se behoshi, dam ghutna (suffocation) aur maut ho sakti hai. CO **colourless aur odourless** hai, is liye mehsoos nahin hoti.

3. Pollution ke sources aur equations

Natural sources: jangal ki aag aur aandhi (dhuan, dust); volcano (dust, zehreeli gases, raakh); deemak aur gaaye (methane); aasmani bijli (nitrogen oxides).

Human activities: coal, petroleum, natural gas jalana. **Taqreeban aadhi air pollution gaariyon se aati hai.**



Dhyan do: kitab ne pehli equation " $C + O_2 \rightarrow CO$ " likhi hai, jo **balanced nahin** (left par 2 O, right par 1). Sahi hai **$2C + O_2 \rightarrow 2CO$** .

4. Ground-level ozone kaise banti hai

Ye **secondary pollutant** hai: kisi jagah se seedhi nahin nikalti, balki doosre pollutants (**nitrogen oxides** aur **VOCs**) ke dhoop mein react karne se banti hai.

Dhoop mein: $\text{NO}_2 \rightarrow \text{NO} + \text{O}$

Phir: $\text{O} + \text{O}_2 \rightarrow \text{O}_3$

Yahan **akela O (atomic oxygen)** sahi hai, kyunki NO_2 toot kar ek oxygen atom deta hai. Baqi har jagah oxygen gas O_2 hi likhni hai.

5. Greenhouse effect aur global warming

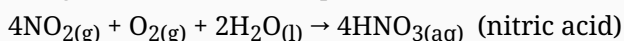
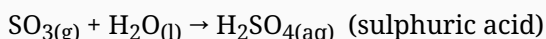
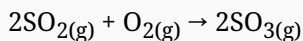
Dhoop zameen par aati hai, zameen garam ho kar **infrared** radiation chhorti hai. CO_2 aur methane jaisi **greenhouse gases** is infrared ko jazb kar ke wapas bhejti hain, **kambal (blanket) ki tarah**, is liye kam گرمی space mein jaati hai. Ye **greenhouse effect** qudrati hai.

Jab insan in gases ko barha deta hai aur zameen zyada garam hoti hai, isay **global warming** kehte hain.

Agar global warming jari rahi: zameen ka temperature barhega; baarish badlegi, kahin **sailaab** kahin **khushk saali**; **polar ice pighlegi** aur samundar ka paani barhega; atmosphere garam ho jayega.

6. Acid rain (sab se zaroori hissa)

Normal baarish ka **pH 5.6** hai. **pH 5.6 se kam** wali baarish **acid rain** hai.



Loha (iron): $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$

Marble: $\text{CaCO}_3 + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$ (2HNO_3 zaroor)

Jaandaron par asar: machhliyan mar jati hain, darakht tabah ho jate hain, jheelein aur daryaa itne acidic ho jate hain ke un mein zindagi nahin reh sakti.

7. Hal (solutions)

Catalytic converter (gaari ke exhaust mein): **platinum, palladium aur rhodium**. CO ko CO_2 , aur NO ko N_2 aur O_2 mein badalta hai.

Acid rain ke liye: kam sulphur wala fuel; power plants mein **flue gas desulphurization** (calcium oxide se SO_2 nikalna); catalytic converters.

Climate ke liye: renewable energy (hawa, dhoop, paani), darakht lagana, electric gaariyan.

Photosynthesis hawa se CO_2 nikalta hai: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Apni hifazat: N95 mask, air quality index dekhna, ghar mein CO detector.

8. Traps: ye pairs ulat mat karna

Ye	Us se farq
Hawa mein oxygen 21%	Nitrogen 78%
CO: haemoglobin se jurti hai	Particulates: lungs mein jate hain, cancer
Greenhouse effect: qudrati kambal	Global warming: insani wajah se zyada گرمی
Normal baarish pH 5.6	Acid rain pH 5.6 se kam
Converter: Pt, Pd, Rh	Calcium oxide: power plant ka flue gas desulphurization
Methane: rice fields, gaaye	NO_x: car engines, bijli; SO₂: coal power plants
Ozone mein O + O₂ (akela O sahi)	Baqi har jagah oxygen gas = O₂

Video (NBF Federal Board, poora Chapter 11):

https://www.youtube.com/watch?v=vovG_r36VOU